

ME 300 Homework 2: Defining Properties and Equilibrium

Answer the following questions neatly on a separate piece of paper or on a tablet. Written answers can be typed. Show all your work and explain your reasoning – partial credit will only be given if you show and explain each of your steps. Submit your homework on Canvas.

- (15 points) What's the difference between a thermodynamic process and a thermodynamic cycle? Define each and describe their differences. How does the state of the system change for each?
- (30 points) To practice thinking about different types of processes, draw the following diagrams for each of the different processes. Label your axes, the two states, and the path between them, just like we did in class.
 - Isothermal process from 1 m^3 to 6 m^3 on a T-V diagram.
 - Isochoric process from 800 K to 400 K on a T-V diagram.
 - Isobaric process from 1200 K to 800 K on a P-T diagram.
 - Isothermal process from 3 MPa to 1 MPa on a P-T diagram.
 - Isobaric process from 3 m^3 to 2 m^3 on a P-V diagram.
 - Isochoric process from 200 kPa to 500 kPa on a P-V diagram.
- (30 points) The concept of quasi-equilibrium is one we'll use on every single analysis throughout the semester. In this problem, we're going to extend our analysis from class on the large ship engines to car engines to determine whether a process can be considered quasi-equilibrium. Mazda is a company that has been extremely innovative in the internal combustion engine space, working to get the most power and highest efficiency of this technology. Their SkyActiv technology is a package that improves engine efficiency significantly while still making the car a lot of fun to drive – do some research online to learn how they do it. In this problem, let's consider the Mazda 3 SkyActiv (hatchbacks for the win) – <https://www.mazdausa.com/vehicles/2022-mazda3-hatchback>.
 - First, explain what the quasi-equilibrium assumption is and why it's important.
 - Next, use the Mazda website to find the bore (width of the cylinder), stroke (length the piston travels over half a rotation), and maximum engine speed (referred to as “redline” on the website) of the engine. Use this information to calculate the timescale of the compression stroke at a highway driving engine speed of 2000 RPM.
 - Use Excel or Matlab to plot the timescale of a compression stroke as a function of engine speed from 0 to the maximum engine speed.
 - At what engine speed would the process no longer be quasi-equilibrium? Why?
- (10 points) Explain why a diatomic molecule like oxygen – O_2 – has a higher specific heat than a noble gas like neon – Ne – in units of J/kmol-K.
- (15 points) For this question, please write a paragraph about what you learned in the Study Skills workshop session you attended. What are three things they talked about that resonated with you and why? These can be things you're doing or things you think might help – explain why these strategies might help you in this course. Finally, explain the growth mindset and how you'll apply it in the context of this course and others you may be taking this summer.